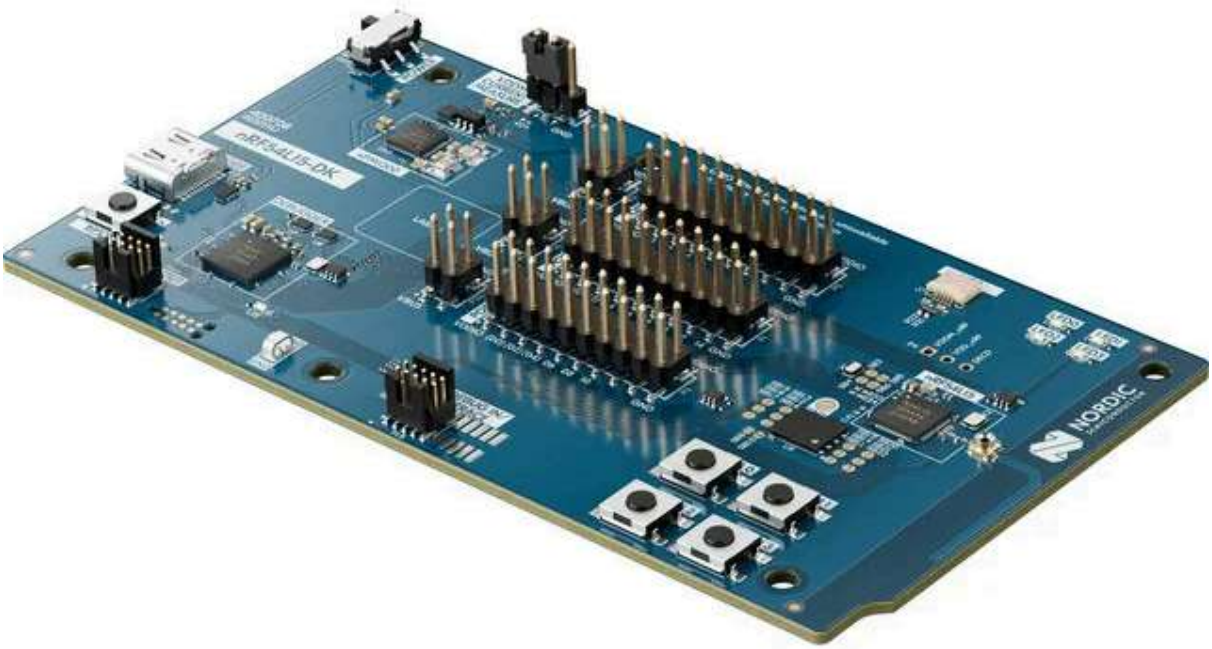
Technical Documentation

: nRF54L15-DK & Bluetooth 6.0 Channel Sounding

Glossary

- **Channel Sounding (CS)**: The breakthrough technology integrated into the nRF54L15 for secure and high-precision distance measurement between two Bluetooth LE devices.
- **Initiator (Requérant)**: The device that requests the distance measurement and typically executes the calculation algorithm.
- **Reflector (Répondeur)**: The target device (e.g., a car key or tracker) that responds to the Initiator's sounding requests.
- **RRAM (Resistive RAM)**: A high-performance non-volatile memory (NVM) technology used in the nRF54L15 SoC, offering up to 1.5 MB of storage.
- **Frequency Hopping**: The rapid switching of radio signals across multiple frequencies (typically utilizing a subset of the 79 available channels) to improve accuracy and reduce interference.
- **Multipath (Trajets multiples)**: A phenomenon where radio signals reach the receiver via multiple paths due to reflections off walls or objects; the nRF54L15 uses up to **4 antenna paths** to mitigate this effect.
- **Zephyr RTOS**: The real-time operating system upon which the nRF Connect SDK is built, providing the foundation for nRF54 series applications.
- **ISM Band (2.4 GHz)**: The Industrial, Scientific, and Medical radio band where Bluetooth 6.0 operates, divided into 79 RF channels.

Hardware Architecture: nRF54L15-DK



The nRF54L15-DK is a versatile development platform designed for ultra-low-power, secure, and highly accurate localization applications.

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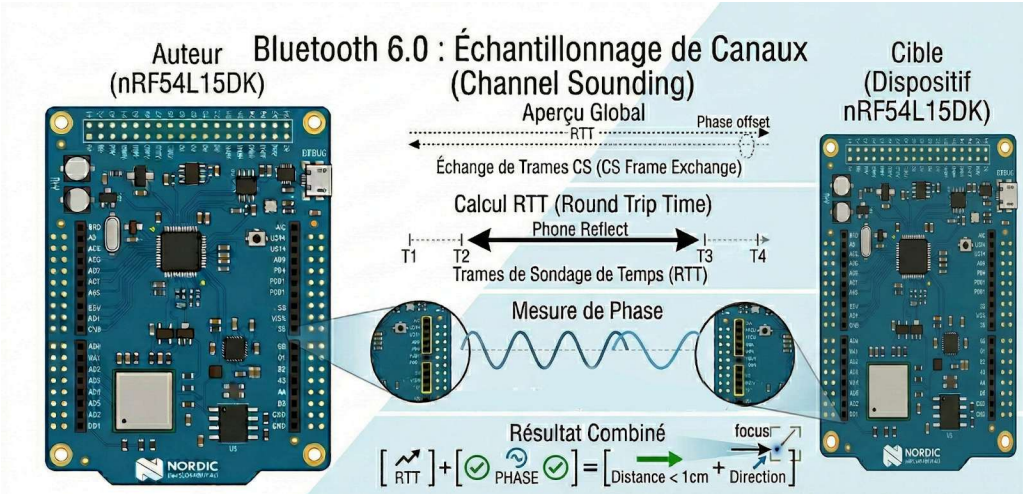
Key Hardware Specifications

- **Processing Core**: Arm® Cortex®-M33 processor clocked at 128 MHz.
- **Memory**: Features up to 1.5 MB of NVM (RRAM) and 256 KB of RAM.
- **Multi-protocol Radio**: Supports Bluetooth 6.0, 802.15.4 (Thread/Matter), and proprietary 2.4 GHz protocols up to 4 Mbps.
- **Debug Interface**: Integrated SEGGER J-Link OB for seamless programming and debugging.
- **Power Consumption**: Extremely low, with 3.2 mA in RX and 5.0 mA in TX at 0 dBm.

Bluetooth 6.0 : Channel Sounding



Bluetooth 6.0 Channel Sounding (CS) represents a paradigm shift from traditional RSSI (signal strength) based ranging. It utilizes the nRF54L15's advanced radio capabilities to perform precise phase and time measurements.

Dual-Layer Measurement Methodology

To achieve a balance between high precision and robust security, the system executes two distinct measurement types during a single CS session:

Phase-Based Ranging (PBR) – High Precision

PBR is the primary method for distance estimation. It calculates the distance by analyzing the phase shift of a radio wave as it travels between the Initiator and the Reflector.

- **The 72-Channel Mechanism:** The nRF54L15 performs rapid Frequency Hopping across a subset of 72 RF channels (within the 79 available in the 2.4 GHz ISM band).
- **Distance Calculation:** By measuring the phase difference relative to the frequency change across these channels, the system solves for distance.
- **Advantage:** This multi-channel approach significantly improves resolution and allows the device to reach sub-meter accuracy (down to 10-30 cm in ideal conditions).

Round-Trip Time (RTT) – Secure Validation

RTT measures the absolute time a packet takes to travel to the target and back.

- **Security Layer:** While PBR is precise, it can be susceptible to sophisticated "manipulation" attacks. RTT acts as a physical security layer; since signals cannot

travel faster than the speed of light, RTT provides a "hard limit" that prevents distance-spoofing (Relay Attacks).

- **UWB Comparison:** This is functionally equivalent to the Two-Way Ranging (TWR) used in UWB systems, providing a coarse but unhackable distance verification.

Signal Integrity & Multipath Mitigation

One of the greatest challenges in indoor localization is Multipath Interference, where signals bounce off walls, floors, and metal objects (Reflected Paths).

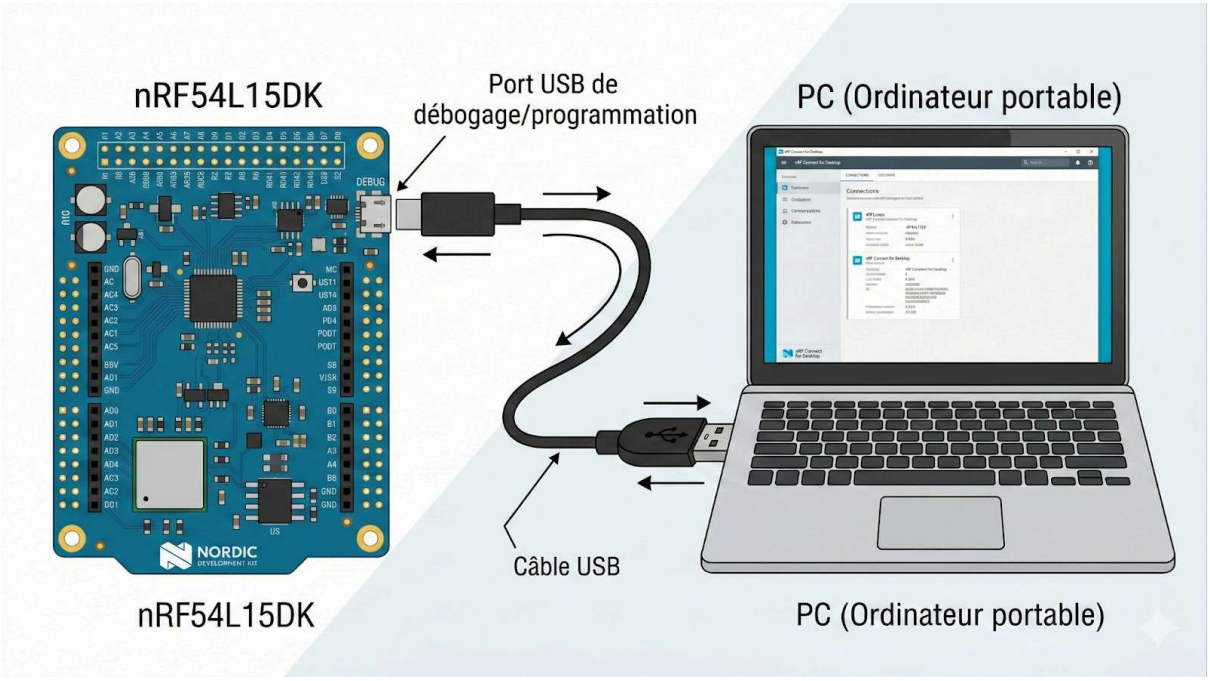
- **Antenna Paths:** The nRF54L15-DK supports up to 4 antenna paths. By switching between different antenna configurations during the 72-channel sweep, the SoC can distinguish the First Path (direct line-of-sight) from later, stronger reflections.
- **Channel Sounding Manager:** The software stack automatically filters out these reflections, ensuring that only the shortest (and therefore correct) distance is reported to the user terminal on the PC.

Roles and Protocol Flow

The communication follows a strict sequence:

1. **Preparation Phase:** Devices agree on the Channel Map (which of the 72 channels to use) and the hopping sequence.
2. **Sounding Phase:** The actual exchange of PBR and RTT packets occurs.
3. **Reflective Feedback:** The *Reflector* sends phase data back to the *Initiator*.
4. **Final Computation:** The *Initiator* (processed by the 128MHz Cortex-M33) combines PBR and RTT data to output a single, secure distance value.

PC to Board Communication



The development workflow requires a physical connection between the DK and a PC for flashing firmware and monitoring real-time distance data.

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1. **Physical Connection**: Connect the board via the USB debug/programming port.
2. **Logical Interface**: The onboard J-Link OB converts USB commands from the PC into signals understandable by the SoC.
3. **Data Feedback**: Distance results and system logs are sent back to the PC via RTT (Real-Time Transfer) or Virtual COM port

Bibliography & Resources

nRF54L15 Product Brief: nordicsemi.com/nRF54L15
Getting Started Guide: https://docs.nordicsemi.com/category/nRF54L15-category